AMA572 Lecture 4.

Leland-Toft Endogenous Default Model

1 Basic Setting of the Model

1.1 Asset Model

Leland and Toft (1996) assume that under a risk-adjusted probability P, the asset value of the firm V_t follows

$$\frac{dV_t}{V_t} = (r - \delta)dt + \sigma dW_t,$$

whose solution is given by

$$V_t = V_0 \exp\left\{(r - \delta - \frac{1}{2}\sigma^2)t + \sigma W_t\right\},$$

with r being the constant risk-free interest rate, δ the total payout rate to the firm's investors (including both bond and equity holders). Note that under the above risk-adjusted probability V_t is a martingale after proper discounting: $V_t = E\left[e^{-(r-\delta)(T-t)}V_T|\mathcal{F}_t\right]$, where $\mathcal{F}_t$ is the information up to time t.

1.2 Debt Issuing and Coupon Payments

The setting of debt issuing and coupon payment follows Leland (1994) and Leland and Toft (1996). In the time interval $(t, t + dt)$, the firm issues new debt with par value pdt, and maturity profile φ, where $\varphi(v) = me^{-mv}$, i.e. the maturity of a specific bond is chosen randomly according to an exponentially distributed random variable with mean $1/m$. At any time t, the par value of debt maturing in the next moment $(s, s + ds)$ is given by

$$\left(\int_{-\infty}^{t} p\varphi(s-u)du\right)ds = \left(\int_{-\infty}^{t} pme^{-m(s-u)}du\right)ds = pe^{-m(s-t)}ds. \quad (1)$$

The above equation (1) has two implications. First, letting $s = t$ in (1) we have at time t the par value of the debt maturing at the next moment $(t, t + dt)$ is given by pdt, which is the same as the par value of the newly-issued debt, achieving a static balance. Secondly, using (1), at time t the par value of all pending debt (with maturity from t to ∞) is a constant, equal to

$$P = \int_{t}^{+\infty} pe^{-m(s-t)}ds = p/m.$$

We also assume that before maturity bondholders will receive coupons at rate ρ until default.

At each moment the firm faces two cash outflows and two cash inflows. The two cash outflows are after-tax coupon payment $(1-\kappa)\rho P dt$ and due principal pdt, where κ is the tax rate; the two cash inflows are $b_t dt$ from selling new debts, where b_t is the price of the total newly issued bonds, and the total payout from the asset $\delta V_t dt$. If the total cash inflow $(\delta V_t + b_t)dt$ is greater than the cash outflow $((1-\kappa)\rho P+p)dt$, we assume that the difference of the two goes to the equity holders as dividends; otherwise, additional equity will be issued to fulfill the due liability. Note that the difference $[((1-\kappa)\rho P + p) - (\delta V_t + b_t)]dt$ is an infinitesimal quantity. Thus, such a financing strategy is feasible as long as the total equity value remains positive before bankruptcy. Therefore, we need to impose the limited liability constraint on the dynamic of the firm's equity.

1.3 Default Payments

As in first passage models, we assume that the default occurs at time $\tau = \inf\{t \geq 0 : V_t \leq V_B\}$. Upon default, the firm loses $(1-\alpha)$ of V_τ, due to reorganization of the firm, and the debtholders as a whole get the rest of the value left, αV_τ, after reorganization.

To find the total debt value, total equity value, and optimal leverage ratio, it is not necessary to spell out the recovery value for individual bonds with different maturities. However, to model credit spreads we need to specify how the remaining asset of the firm αV_τ is distributed among bonds with different maturities. Three standard assumptions are recovery at a fraction of par value, of market values, and of corresponding treasury bonds. Here we shall use the assumption of recovery at a fractional treasury bonds. We use this assumption because it makes analytical calculation easier. More precisely, upon default the payoff of a bond with unit face value and maturity $T > \tau$ is given by

$$ce^{-r(T-\tau)}, \;\; 0 \leq c \leq 1.$$

To determine c, we shall match the total payment to bondholder with the total remaining asset αV_τ. By the memoryless property of exponential distribution, the conditional distribution of a bond's maturity does not change given its maturity is beyond τ. Thus, we have by (1)

$$\int_\tau^{+\infty} ce^{-r(s-\tau)} \cdot pe^{-m(s-\tau)} ds = \alpha V_\tau,$$

yielding

$$c = \frac{m+r}{m}\frac{\alpha V_\tau}{P} = \frac{m+r}{m}\frac{\alpha V_B}{P}.$$

To make sure that $0 \leq c \leq 1$, we shall impose that

$$\frac{m+r}{m}\frac{\alpha V_B}{P} \leq 1. \tag{2}$$

We will see that this assumption is satisfied by the optimal default barrier; see Theorem 1.

We should emphasize that this recovery assumption is irrelevant if one is only interested in understanding debt values as a whole, but is relevant when we consider credit spreads, which in turn depend on cash flows to bonds with different maturities. In particular, our results about optimal default boundary and optimal leverage level will not be affected by the recovery assumption.

1.4 Debt, Equity, and Market Value of the Firm

At time 0 the price of a bond with face value 1 and maturity T when the firm asset $V_0 = V$ is given by

$$\begin{aligned}
B(V;V_B,T) &= E\left[e^{-rT}\mathbf{1}_{\{\tau>T\}} + e^{-r\tau}\cdot ce^{-r(T-\tau)}\mathbf{1}_{\{\tau\leq T\}}\right] + E\left[\int_0^{\tau\wedge T}\rho e^{-rs}ds\right] \\
&= e^{-rT}P[\tau\geq T] + cE[e^{-rT}\mathbf{1}_{\{\tau\leq T\}}] + \frac{\rho}{r}(1-E[e^{-r(\tau\wedge T)}]) \\
&= \left(1-\frac{\rho}{r}\right)e^{-rT}P[\tau\geq T] + cE\left[e^{-rT}\mathbf{1}_{\{\tau\leq T\}}\right] + \frac{\rho}{r}\left(1-E\left[e^{-r\tau}\mathbf{1}_{\{\tau\leq T\}}\right]\right).
\end{aligned}$$

In our endogenous model, both P (which is related to the optimal debt/equity ratio) and the endogenous default barrier V_B will be decision variables.

According to the Modigliani-Miller theorem, the total market value of the firm is the asset value plus tax benefits and minus the bankruptcy costs. More precisely, the market value of the firm at time 0 is

$$\begin{aligned}
v(V;V_B) &= V + E\left[\int_0^{\tau}\kappa\rho P e^{-rt}dt\right] - (1-\alpha)E[V_\tau e^{-r\tau}] \\
&= V + \kappa\rho P\cdot E\left[\int_0^{\tau}e^{-rt}dt\right] - (1-\alpha)V_B E[e^{-r\tau}].
\end{aligned}$$

Note that we have to put a constraint on ρ,

$$\frac{r}{\kappa} > \rho; \tag{3}$$

otherwise, the total tax benefit, $E[\int_0^\tau \kappa\rho P e^{-rt}dt] \leq \kappa\rho P\int_0^\infty e^{-rt}dt = \frac{\kappa\rho}{r}P$, could exceed the face value of the debt P, which may create a tax abitrage

opportunity. The total equity value equals to the total value of the firm less the total value of bonds,

$$S(V;V_B) = v(V;V_B) - D(V;V_B),$$

where the total debt $D(V;V_B)$ is equal to

$$D(V;V_B) = P\int_0^{+\infty} me^{-mT}B(V;V_B,T)dT.$$

We should point out that in this static model (thanks to the exponential maturity profile), the total firm value, total debt and equity values are all Markovian, and are independent of the time horizon.

1.5 Preliminary Results for Debt and Equity Values

Similarly the total market value of the firm $v(V;V_B)$ can be computed similarly in a hwk problem, and the total equity of the firm is given by $S(V;V_B) = v(V;V_B) - D(V;V_B)$.

Proposition 1. The total market value of the firm is given by

$$v(V;V_B) = V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right](V_B/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi}. \quad (4)$$

The total debt value of the firm is given by

$$D(V;V_B) = P\frac{\rho+m}{r+m} - P\left(\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right)(V_B/V)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi}. \quad (5)$$

The total equity value of the firm is given by

$$\begin{aligned} S(V;V_B) &= v(V;V_B) - D(V;V_B) \\ &= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right](V_B/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi} - P\frac{\rho+m}{r+m} \\ &\quad + P\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right](V_B/V)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi}. \end{aligned} \quad (6)$$

Proof. The market value of the firm at time 0 is

$$\begin{aligned} v(V;V_B) &= V + \kappa\rho P \cdot E\left[\int_0^\tau e^{-rt}dt\right] - (1-\alpha)V_B E[e^{-r\tau}] \\ &= V + \kappa\rho P \cdot E\left[\frac{1}{r}(1-e^{-r\tau})\right] - (1-\alpha)V_B E[e^{-r\tau}] \\ &= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\cdot E[e^{-r\tau}]. \end{aligned}$$

Since

$$E[e^{-r\tau}] = \exp\left\{\log(V_B/V)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\right\},$$

where $\xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2}$, we must have (4).

We know from the previous discussions that the Laplace transform of a bond price, $B(V;V_B,T)$, is given by

$$\begin{aligned}&\int_0^\infty e^{-\beta T}B(V;V_B,T)dT\\ &= \frac{\rho+\beta}{\beta(r+\beta)} - \left(\frac{\rho}{r}\frac{1}{\beta} + \left(1-\frac{\rho}{r}-c\right)\frac{1}{r+\beta}\right)\cdot\exp\left\{\log(V_B/V)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}(\beta+r)}+\xi\right)\right\},\end{aligned}$$

where

$$\xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2}, \qquad c = \frac{m+r}{m}\frac{\alpha V_B}{P}.$$

Therefore, the value of total debt at time 0 is

$$\begin{aligned}D(V;V_B) &= P\int_0^{+\infty} me^{-mT}B(V;V_B,T)dT\\ &= P\frac{\rho+m}{r+m} - P\left(\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right)\\ &\quad\times\exp\left\{\log(V_B/V)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}(m+r)}+\xi\right)\right\}\\ &= P\frac{\rho+m}{r+m} - P\left(\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right)\cdot(V_B/V)^{\sqrt{\xi^2+2(m+r)/\sigma^2}+\xi},\end{aligned}$$

which is (5). The combination of equation (4) and (5) yields that the total equity value of the firm is given by (6).

Remark: Note that we are able to compute the total debt value, total equity value, and the total market value analytically, although we can only compute a given bond price via Laplace transforms. This is made possible thanks to our assumption of the exponential maturity profile.

2 Optimal Capital Structure and Endogenous Default

In this section, we consider the problem of optimal capital structure and optimal endogenous default barrier, i.e. the choices of optimal debt level P and bankrupt trigger V_B.

2.1 The Solution of a Two-Stage Optimization Problem

Deciding optimal P and choosing optimal V_B are two entangled problems, and cannot be separated easily. For example, when a firm chooses P to maximize the total firm value at time 0, the decision on P obviously depends on the bankruptcy trigger level V_B. Similarly, after the debt being issued, the equity holders will choose an optimal V_B, which of course depends on how much debt the firm has. Here we shall choose P and V_B according to a two-stage optimization procedure as in Leland (1994) and Leland and Toft (1996).

More precisely, for a fixed P, the equity holders find the optimal default barrier by maximizing the equity value, subject to the limited liability constraint. It is the equity value that is maximized, because after the debt being issued it is the equity holders who control the firm. Mathematically, the equity holders find the best V_B by solving

$$\max_{V_B} S(V; V_B) \tag{7}$$

subject to the limited liability constraint

$$S(V'; V_B) \geq 0, \text{ for all } V' \geq V_B \geq 0. \tag{8}$$

The limited liability constraint is imposed so that the equity value is always nonnegative for any future firm asset value V', as long as the asset value V' is above the bankruptcy level V_B. The is the first stage optimization problem.

Clearly the optimal $V_B^* \equiv V_B^*(P)$ to be chosen by the equity holders will depend on P. At time 0, the firm will conduct the second stage optimization to maximize the firm value, in anticipation of what the equity holders will do later. More precisely, the second stage optimization is that

$$\max_{P} v(V; V_B^*(P)). \tag{9}$$

The two-stage optimization problem partly arises due to the conflict of interests between debt and equity holders. It is obvious that choosing the leverage level P and bankrupt trigger V_B simultaneously will lead to a better market value of the firm. Leland (1998) uses the difference of the two to explain agency costs due to conflicts between the equity and debt holders. Obviously the two-stage optimization is only a rough approximation to complicated decision and negotiation between debt and equity holders. Here is the result for the two-stage optimization.

Theorem 1. (a) The solution of

$$\frac{\partial S(V; V_B^*)}{\partial V}|_{V=V_B^*} = 0 \tag{10}$$

is given by $V_B^* = \epsilon P$, where

$$\epsilon = \frac{\frac{\rho+m}{r+m}\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right) - \frac{\kappa\rho}{r}\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)}{1+(1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)+\alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)}.$$

(b)

$$0 < \epsilon < \frac{1}{\alpha}\frac{\rho+m}{r+m}, \tag{11}$$

and, therefore, the default barrier V_B^* satisfies (2).

(c) The first stage optimization: Given the debt level P, the optimal barrier level solving (7) with the constraint in (8) is given by $V_B^* = \epsilon P$.

(d) The second stage optimization: After plugging the optimal $V_B^* = \epsilon P$ into the second stage optimization, we have that $v(V; \epsilon P)$ is a concave function of $P \in (0, V/\epsilon)$, which implies that we can find a unique optimal debt level P for the problem (9).

Proof. (a) It follows from equation (6) that

$$\begin{aligned}
&\frac{\partial S(V;V_B)}{\partial V} \\
&= 1 - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi-1}\left(-\frac{V_B}{V^2}\right) \\
&\quad + P\left[\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi-1}\left(-\frac{V_B}{V^2}\right).
\end{aligned}$$

Hence,

$$\begin{aligned}
&\frac{\partial S(V;V_B)}{\partial V}\Big|_{V=V_B} \\
&= 1 - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\left(-\frac{1}{V_B}\right) \\
&\quad + P\left[\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\left(-\frac{1}{V_B}\right).
\end{aligned}$$

Letting

$$\frac{\partial S(V;V_B)}{\partial V}\Big|_{V=V_B} = 0,$$

we obtain

$$1+\frac{\kappa\rho}{r}P\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\frac{1}{V_B}+(1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)$$
$$-P\left[\frac{\rho}{r}+\left(1-\frac{\rho}{r}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\frac{1}{V_B}+\alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)=0,$$

which yields

$$V_B^*=\frac{\left[\frac{\rho}{r}+\left(1-\frac{\rho}{r}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)-\frac{\kappa\rho}{r}\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)}{1+(1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)+\alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)}P.$$

So $V_B^*=\epsilon P$.

(b) See a homework problem.

(c) This part is the most difficult part; see the paper by Chen and Kou (2009).

(d) By equation (4) we have

$$\begin{aligned}v(V;\epsilon P)&=V+\frac{\kappa\rho}{r}P-\left[\frac{\kappa\rho}{r}P+(1-\alpha)\epsilon P\right](\epsilon P/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi}\\&=V+\frac{\kappa\rho}{r}P-\left[\frac{\kappa\rho}{r}+(1-\alpha)\epsilon\right](\epsilon/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi}\cdot P^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi+1}.\end{aligned}$$

Since

$$\sqrt{\xi^2+\frac{2r}{\sigma^2}}+\xi+1>\sqrt{\xi^2}+\xi+1\geq 1,$$

it follows that

$$P^{\left(\sqrt{\xi^2+\frac{2r}{\sigma^2}}+\xi+1\right)}\text{ is convex,}$$

and consequently

$$-\left[\frac{\kappa\rho}{r}+(1-\alpha)\epsilon\right]\left(\frac{\epsilon}{V}\right)^{\sqrt{\xi^2+\frac{2r}{\sigma^2}}}\cdot P^{\sqrt{\xi^2+\frac{2r}{\sigma^2}+\xi+1}}.$$

The expression above is concave.

Since $V+\frac{\kappa\rho}{r}P$ is linear and the sum of a linear function and a concave function is still concave, it follows that the function $v(V;\epsilon P)$ is concave.

2.2 The Optimal Capital Structure

Table 2 shows the effect of various parameters on the optimal leverage level. The parameters are $\kappa = 35\%$, $r = 8\%$, $\rho = 8.162\%$, $\delta = 6\%$, $V_0 = 100$. The riskless rate $r = 8\%$ is close to the average historical short-term treasury rate during 1973-1998, and the coupon rate $\rho = 8.162\%$ is the par coupon rate for risk-free bonds with semi-annual coupon payments when the continuously compounded interest rate is 8%.

Consistent with our intuition, the table shows that the optimal leverage ratio is an increasing function of the fractional remaining asset α and average maturity profile $1/m$, and is a decreasing function of diffusion volatility σ.

	$m^{-1} = 0.5$		$m^{-1} = 1$		$m^{-1} = 2$		$m^{-1} = 5$	
	$\sigma = 0.2$	$\sigma = 0.4$	$\sigma = 0.2$	$\sigma = 0.4$	$\sigma = 0.2$	$\sigma = 0.4$	$\sigma = 0.2$	$\sigma = 0.4$
$\alpha = 5\%$	7.14%	1.11%	11.19%	2.37%	17.56%	5.04%	30.67%	12.91%
$\alpha = 25\%$	13.78%	3.21%	18.31%	5.28%	25.08%	9.13%	38.41%	19.07%
$\alpha = 50\%$	25.44%	9.22%	30.30%	12.63%	37.29%	18.27%	50.52%	31.19%

However, the Leland-Toft model cannot explain why many high-tech companies have no debt at all.

HWK

1. Prove (11).

AMA572 Lecture 5

Intensity Based Reduced Form Models and Introduction to Credit Derivatives

Unlike previous structural models, such as Merton, Black-Cox, and Leland-Toft models, which assume that there is a fundamental asset value which triggers the default, reduced form models do not model default explicitly. More precisely, reduced form models assume that there is some exogenous processes, such as intensity processes, drive the defaults, and take the bond prices as given. After we calibrate the defaultable bond price to the market, then we can use the bonds prices to compute prices for other derivatives. This is similar to the Black-Scholes model, in which we use stock prices to compute option prices.

Popular reduced form models include Duffie-Singleton model, which is based on intensities, and Jarrow-Turnbull model, which is based on credit rating migration. Here we shall focus on the first model.

1 Duffie-Singleton Reduced Form Model

Reduced-form models take the probability of default as given by the market, unlike the structural models (such as Merton, Black-Cox, Leland-Toft models). In this lecture we shall focus on one particular reduced-form model, Duffie-Singleton (1999) Model.

1.1 Hazard Rate and Intensity

The hazard rate $h(t,T)$ measures the conditional probability of defaulting between T and $T+dt$, given all the information available at time t. More precisely, let the default time be τ

$$S(t,T) = P(\tau > T|\mathcal{F}_t),$$

$$f(t,T) = \frac{\partial S(t,T)}{\partial T},$$

$$h(t,T) = \frac{f(t,T)}{S(t,T)} = \frac{1}{S(t,T)}\frac{\partial S(t,T)}{\partial T}.$$

Then, it can be shown that

$$S(t,T) = \exp\left\{-\int_t^T h(t,s)ds\right\}.$$

The intensity $\lambda(t)$ measures the instantaneous defaults rate. More precisely, there is a unique stochastic process $\lambda(t)$ such that

$$N(t) - \int_0^t \lambda(s)ds$$

is a martingale, and under suitable conditions

$$\lambda(t) = h(t,t).$$

Here $N(t)$ is a counting process

$$N(t) = I\{\tau \leq t\}.$$

1.2 The Model

Consider a European contingent claim whose payoff is given by $g(T)$. This is a quite general framework. For example, for the zero coupon bond $g(T) = 1$. Suppose the recovery value is a fraction, δ, of the market value of the contingent claim right before the default. The key result of Duffie-Singleton model is that the value of the contingent claim at time t is given by

$$g(t) = E^*\left[\exp\left\{-\int_t^T [r(s) + (1-\delta)\lambda(s)]\, ds\right\} \cdot g(T)|\mathcal{F}_t\right]. \qquad (1)$$

Remarks. (1) In the above formula the risk-free rate $r(s)$ may be stochastic, and the default intensity may also be a stochastic process. (2) For zero-coupon bond we have at time 0

$$B(0,T) = E^*\left[\exp\left\{-\int_0^T [r(s) + (1-\delta)\lambda(s)]\, ds\right\}\right].$$

(3) After we calibrate the default intensity $\lambda(t)$, then we can use $\lambda(t)$ to price other contingent claims. In other words, we view bond prices as given, and then use the bond prices to compute prices for other derivatives. This is similar to the Black-Scholes model, in which we use stock prices to compute option prices.

1.3 A Heuristic Derivation

We shall only give a heuristic explanation of (1). Consider a discrete time framework. We can think $h(t)dt$ as approximately the default probability in $(t, t+dt)$ and $r(t)$ as the approximate risk-free yield from t to $t+dt$. Then the standard option pricing theory implies that

$$g(t) = h(t)dt \cdot e^{-r(t)dt} E^*[\delta g(t+dt)|\mathcal{F}_t] + (1 - h(t)dt)e^{-r(t)dt} E^*[g(t+dt)|\mathcal{F}_t],$$

because the default payment is $\delta g(t+dt)$ and the non-default payment is $g(t+dt)$.

Introduce $R(t)$ such that

$$\exp\{-R(t)\cdot dt\} = h(t)dt\cdot e^{-r(t)dt}\delta + (1-h(t)dt)e^{-r(t)dt}. \tag{2}$$

Then

$$g(t) = \exp\{-R(t)\cdot dt\}E^*[g(t+dt)|\mathcal{F}_t].$$

Iterating the above expression yields

$$g(t+dt) = \exp\{-R(t+dt)\cdot dt\}E^*[g(t+2dt)|\mathcal{F}_{t+dt}],$$

and therefore

$$\begin{aligned} g(t) &= \exp\{-R(t)\cdot dt\}E^*[g(t+dt)|\mathcal{F}_t] \\ &= \exp\{-R(t)\cdot dt\}\cdot E^*\left\{\exp\{-R(t+dt)\cdot dt\}\cdot E^*[g(t+2dt)|\mathcal{F}_{t+dt}]|\mathcal{F}_t\right\} \\ &= E^*\left\{E^*[\exp\{-\left(R(t)+R(t+dt)\right)\cdot dt\}\cdot g(t+2dt)|\mathcal{F}_{t+dt}]|\mathcal{F}_t\right\} \\ &= E^*[\exp\{-\left(R(t)+R(t+dt)\right)\cdot dt\}\cdot g(t+2dt)|\mathcal{F}_t]. \end{aligned}$$

Thus, continuing this procedure, we have

$$g(t) = E^*[\exp\{-\left(R(t)+R(t+dt)+\cdots+R(T)\right)\cdot dt\}\cdot g(T)|\mathcal{F}_t]. \tag{3}$$

Now we need to have an approximation for $R(t)$. Using the approximation $e^x \approx 1+x$, we have

$$\exp\{-R(t)\cdot dt\} \approx 1 - R(t)\cdot dt,$$

$$\lambda(t)dt\cdot e^{-r(t)dt}\delta + (1-\lambda(t)dt)e^{-r(t)dt} \approx \lambda(t)dt\cdot(1-r(t)dt)\delta + (1-\lambda(t)dt)(1-r(t)dt),$$

and, therefore, we have from (2) that

$$1 - R(t)\cdot dt \approx \lambda(t)dt\cdot(1-r(t)dt)\delta + (1-\lambda(t)dt)(1-r(t)dt).$$

In summary, we have

$$1 - R(t)\cdot dt \approx \lambda(t)\delta dt - \lambda(t)r(t)\delta\left(dt\right)^2 + 1 - \lambda(t)dt - r(t)dt + \lambda(t)r(t)\left(dt\right)^2.$$

Ignoring the term $(dt)^2$ we have

$$-R(t)\cdot dt \approx \lambda(t)\delta dt - \lambda(t)dt - r(t)dt,$$

or

$$R(t) \approx \lambda(t)(1-\delta) + r(t).$$

This in conjunction with (3) gives, as $dt \to 0$,

$$\begin{aligned} g(t) &= E^*\left[\exp\{-\int_t^T R(s)ds\}\cdot g(T)|\mathcal{F}_t\right] \\ &= E^*\left[\exp\{-\int_t^T \left[\lambda(s)(1-\delta)+r(s)\right]ds\}\cdot g(T)|\mathcal{F}_t\right], \end{aligned}$$

which shows the result.

2 Credit Default Swaps

2.1 Background

A credit default swap (CDS) is a traded insurance contract that provides payments against the risk of default. The buyer of the CDS makes periodic payment to the seller until either the contract reaches maturity or a credit event occurs. In the credit event, the seller of the CDS will make certain payment. For example, a buyer may pay 90 basis point on a notation amount $\$10M$ once per year (i.e. \$90K per year) for 5 years, in exchange for a protection payment of difference between the face value of the debt and the market value of the debt after the credit event. For example, if the face value of the debt is \$100, but the market value is \$40 after the credit event, and the seller of the CDS will pay \$6m to the buyer.

Back in 2008, all CDS contracts are not traded in exchanges. Now CDS contracts are traded in several exchanges. Even individual investors can buy CDS contracts via, e.g., International Exchange (ICE) and Chicago Mercantile Exchange (CME).

Two popular indices for CDS are: 5 year and 10 year CDX NA IG index which tracks 125 investment grade North American companies; 5 year and 10 year iTraxx Europe index which tracks 125 investment grade European companies.

In September 2009 US government bailed American International Group (AIG) out, mainly because AIG had been excessively selling CDS protection without hedging against the possibility that the reference entities, especially on Lehman, might decline in value, which exposed the insurance giant to potential losses over \$100 billion.

In April 2012, the market in credit default swaps was possibly being affected by the activities of Bruno Iksil, a trader for J.P. Morgan Chase & Co., referred to as "the London whale" in reference to the huge positions he was taking. Heavy opposing bets to his positions were made by traders, including another branch of J.P. Morgan, who purchased the derivatives offered by J.P. Morgan in such high volume. \$2 billion losses were reported by the firm in May, 2012 in relationship to these trades.

2.2 Valuation

We will illustrate the valuation of 5 year CDS via an example. In this example, we shall find out the break-even CDS spread s.

Assume:

1. The default intensity during any year is 0.02.

2. The risk-free rate is 5%

3. The recovery rate is 40%.

4. Default can only happen at year 0.5, 1.5, 2.5, 3.5, 4.5.

Since the payment from the buyer is typically in arrear (e.g. at year 1, 2, ..., 5), the calculation relies on computation of 3 payments.

1. Regular payments by the buyer.

2. Accrual payment by the buyer after the credit event.

3. The payment from the seller to the buyer after the credit event.

The following table details item 1, with s being the CDS spread.

Time (Years)	Prob. of Survival	Expected Payments	Discount Factor	PV of Exp Payment
1	0.9800	$0.9800s$	0.9512	$0.9322s$
2	0.9604	$0.9604s$	0.9048	$0.8690s$
3	0.9412	$0.9412s$	0.8607	$0.8101s$
4	0.9224	$0.9224s$	0.8187	$0.7552s$
5	0.9039	$0.9039s$	0.7788	$0.7040s$
Total				$4.0704s$

For example, in Year 3, the survival probability is

$$(1 - 0.02)^3 = 0.9412;$$

the discounted factor is

$$e^{-0.05*3} = 0.860\,7;$$

and the PV of expected payment is

$$0.9412s * 0.860\,7 = 0.810\,1s.$$

The following table details item 2, with s being the CDS spread.

Time (Years)	Prob. of Default	Expected Payments	Discount Factor	PV of Exp Payment
0.5	0.0200	$0.0100s$	0.9753	$0.0097s$
1.5	0.0196	$0.0098s$	0.9277	$0.0091s$
2.5	0.0192	$0.0096s$	0.8825	$0.0085s$
3.5	0.0188	$0.0094s$	0.8395	$0.0079s$
4.5	0.0184	$0.0092s$	0.7985	$0.0074s$
Total				$0.0426s$

For example, the default probability during the third year (which is assumed to happen exactly at time 2.5 year) is

$$0.02 * (1 - 0.02)^2 = 0.0192;$$

and the corresponding arrear payment is

$$(2.5 - 2) * 0.0192s = 0.009\,6s.$$

The discount factor is

$$e^{-0.05*2.5} = 0.882\,50.$$

The following table details item 3, with s being the CDS spread.

Time (Years)	Prob. of Default	Expected Payments	Discount Factor	PV of Exp Payment
0.5	0.0200	0.0120	0.9753	0.0117
1.5	0.0196	0.0118	0.9277	0.0109
2.5	0.0192	0.0115	0.8825	0.0102
3.5	0.0188	0.0113	0.8395	0.0095
4.5	0.0184	0.0111	0.7985	0.0088
Total				0.0511

For example, in the above table, if the default happens at time 2.5, then the expected payment is

$$0.0192 * (1 - 0.4) = 0.011\,5.$$

The break-even CDS spread is solved by matching the payments between the buyer and seller, i.e.,

$$4.0704s + 0.0426s = 0.0511,$$

which yields

$$s = 1.24\%,$$

or 124 basis points per year, i.e. \$124,000 per year on the \$10M contact.

Note (i) Market CDS spread may be different from the break-even CDS spread, which may due to the different default intensities being used. In this case, traders may want to enter a trade, either on the long or short side of CDS. (2) Even if the traders agree on the initial CDS spread at time 0, at time t, the CDS spread may be considerably different from the spread at time 0, resulting in a profit/loss for one party.

HWK:

1. Suppose that the risk-free rate is at 7% per year with continuous compounding and that defaults can occur half way through each year in a new 5-year credit default swap. Suppose that the recovery rate is 30% and the default intensity each year is 3%. What is the break-even spread for the credit default swap? Assume payments are made annually.